

Tsz-Kin (Eric) Lai

Research Scientist

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Research Interests

Tropical Cyclones, Mesoscale Convective Systems, Tropical Meteorology, Severe Weather, Cloud Physics, Numerical Weather Prediction

Education

Ph.D. in Atmospheric and Oceanic Sciences

2021

McGill University

Montreal, Canada

- Thesis: *Impacts of Asymmetric Dynamics on Tropical Cyclone Eyewall Replacement Cycles*
- Supervisor: Prof. M. K. (Peter) Yau
- CMOS' Tertia M.C. Hughes Memorial Graduate Student Prize for the Excellent Thesis in 2021

M.Sc. in Atmosphere, Ocean and Climate

2013

University of Reading

Reading, UK

- Dissertation: *Emergence of Tropical Cyclones in Baroclinic Waves*
- Supervisors: Prof. John Methven and Prof. Rosalind J. Cornforth
- Graduated with Distinction

B.Sc. in Physics with minor in Earth System Science

2012

The Chinese University of Hong Kong

Hong Kong

- Bachelor Final Year Project: *Advanced Bias Removal Approach Using a Kalman Filter for Probabilistic Wind Speed Forecasts During the Periods of Tropical Cyclone Influence*
- Supervisors: Mr. Ping Cheung (Hong Kong Observatory) and Dr. Kam-Moon Pang

Research and Work Experience

Research

Research Scientist

Aug 2024–Present

Environment and Climate Change Canada

Dorval, Canada

Postdoctoral Research Associate

Dec 2021–Jul 2024

Space and Atmospheric Physics Group, Department of Physics, Imperial College London

- Advisor: Prof. Ralf Toumi

Postdoctoral Researcher

Jul 2021–Dec 2021

Department of Atmospheric and Oceanic Sciences, McGill University

Montreal, Canada

- Advisor: Prof. M. K. (Peter) Yau

Visiting Graduate Student

Jul 2019–Oct 2019

Research Applications Laboratory (RAL), National Center for Atmospheric Research (NCAR)

Boulder, USA

- Tropical Cyclone Inner Eyewall Decay in Numerical Experiments
- Host: Dr. Eric A. Hendricks

Graduate Research Assistant

Sep 2015–Jun 2021

Department of Atmospheric and Oceanic Sciences, McGill University

Montreal, Canada

Meteorology Researcher Dec 2013–Jul 2015
 Fugro GEOS Ltd. (now Fugro GB Marine Ltd.) South Oxfordshire, UK

- Participated in a meteorological research project “IFADS: Improving Forecasts of African Dust Storms” in collaboration with University of Leeds
- Participated in a research project on “Holistic Vessel Performance and Routing System” in collaborations with University of Southampton etc.
- Performed internal projects such as Nigeria lightning statistics and validation of wind speed forecasts produced by WRF and GFS

Voluntary Research Assistant Aug 2013–Aug 2015
 Department of Meteorology, University of Reading Reading, UK

- Performed a research project about tropical cyclogenesis based on my master’s dissertation

Summer Intern Student Summer 2011
 Hong Kong Observatory Hong Kong

- Worked on “Improving the very short range convective weather forecast for the Hong Kong Flight Information Region (HKFIR)”
- Developed an improved forecasting approach based on ECMWF model data

Student Research Assistant Summer 2010
 Institute of Space and Earth Information Science, The Chinese University of Hong Kong Hong Kong

- Studied the influence of ENSO on the weather in Hong Kong

Teaching and Technical.....

Teaching Assistant Sep 2016–Apr 2020
 Department of Atmospheric and Oceanic Sciences, McGill University Montreal, Canada

- ATOC214 Introduction: Physics of the Atmosphere (Fall 2016, Fall 2018)
- ATOC184 Science of Storms (Winter 2017, Winter 2018, Winter 2019, Winter 2020)
- ATOC181 Introduction to Atmospheric Science (Fall 2017)

Technical Helper for Community Weather Information Network (Co-WIN) Summer 2009
 Department of Applied Physics, The Hong Kong Polytechnic University Hong Kong

- Monitored, maintained and upgraded the automatic weather stations of Co-WIN
- Performed data testing and apparatus calibration

Selected Honours and Awards

2022: Tertia M.C. Hughes Memorial Graduate Student Prize for the Excellent Thesis in 2021,
Canadian Meteorological and Oceanographic Society (CMOS), Canada

2019: Graduate Mobility Award, *McGill University, Canada*

2015–2018: Graduate Excellence Fellowship, *McGill University, Canada*

2015–2016: Atmospheric and Oceanic Sciences Graduate Award, *McGill University, Canada*

2012: International Masters Bursary, *University of Reading, UK*

2011: Hong Kong Observatory Scholarship, *Hong Kong Observatory, Hong Kong*

2011: Second Runner-up in Undergraduate Individual Entry, Professor Sir Charles K. Kao Student Creativity Awards 2011, *The Chinese University of Hong Kong, Hong Kong*

Peer-Reviewed Publications

Published.....

- Lai, T.-K.,** and H. Lin, 2026: Subseasonal variability of North American tropical cyclone landfalls and phase transitions over the Atlantic coasts. *npj Nat. Hazards*, **3**, 55, doi:10.1038/s44304-026-00202-x.

- Ito, K., Y. Miyamoto, C.-C. Wu, A. Didlake, J. Hlywiak, Y.-H. Huang, **T.-K. Lai**, L. Pattie, N. Qin, U. Shimada, D. Tao, Y. Yamada, J. A. Zhang, S. Kanada, and D. Herndon, 2025: Recent research and operational tools for improved understanding and diagnosis of tropical cyclone inner core structure. *J. Meteor. Soc. Japan*, **103**, 147–180, doi:10.2151/jmsj.2025-008.
- **Lai, T.-K.**, and R. Toumi, 2025: Sensitivity of the energy conversion efficiency of tropical cyclones during intensification to sea surface temperature and static stability. *Quart. J. Roy. Meteor. Soc.*, **151**, e4895, doi:10.1002/qj.4895.
- **Lai, T.-K.**, and R. Toumi, 2023: Has there been a recent shallowing of tropical cyclones?. *Geophys. Res. Lett.*, **50**, e2022GL102184, doi:10.1029/2022GL102184.
- **Lai, T.-K.**, E. A. Hendricks, and M. K. Yau, 2021: Long-term effect of barotropic instability across the moat in double-eyewall tropical cyclone-like vortices in forced and unforced shallow-water models. *J. Atmos. Sci.*, **78**, 4103–4126, doi:10.1175/JAS-D-21-0065.1.
- **Lai, T.-K.**, E. A. Hendricks, M. K. Yau, and K. Menelaou, 2021: Roles of barotropic instability across the moat in inner eyewall decay and outer eyewall intensification: Essential dynamics. *J. Atmos. Sci.*, **78**, 1411–1428, doi:10.1175/JAS-D-20-0169.1.
- **Lai, T.-K.**, E. A. Hendricks, K. Menelaou, and M. K. Yau, 2021: Roles of barotropic instability across the moat in inner eyewall decay and outer eyewall intensification: Three-dimensional numerical experiments. *J. Atmos. Sci.*, **78**, 473–496, doi:10.1175/JAS-D-20-0168.1.
- **Lai, T.-K.**, K. Menelaou, and M. K. Yau, 2019: Barotropic instability across the moat and inner eyewall dissipation: A numerical study of Hurricane Wilma (2005). *J. Atmos. Sci.*, **76**, 989–1013, doi:10.1175/JAS-D-18-0191.1.
- Menelaou, K., M. K. Yau, and **T.-K. Lai**, 2018: A possible three-dimensional mechanism for oscillating wobbles in tropical cyclone-like vortices with concentric eyewalls. *J. Atmos. Sci.*, **75**, 2157–2174, doi:10.1175/JAS-D-18-0005.1.

Invited Talk

- **Lai, T.-K.**, E. A. Hendricks, K. Menelaou, and M. K. Yau, 2019: Barotropic instability across the moat and inner eyewall dissipation: A real case simulation and numerical experiments. *NCAR MMM Dynamics Happy Hour Seminar Series*, Boulder, CO, USA.

Recent Conference Presentations

- **Lai, T.-K.**, and H. Lin, 2026: Subseasonal variability of North American tropical cyclone landfalls and phase transitions over the Atlantic coasts. *AMS 37th Conference on Hurricanes and Tropical Meteorology*, San Diego, CA, USA. (Oral)
- **Lai, T.-K.**, and R. Toumi, 2024: Observed trend of inner-core depth and simulated energy conversion efficiency of tropical cyclones. *AMS 36th Conference on Hurricanes and Tropical Meteorology*, Long Beach, CA, USA. (Oral)
- **Lai, T.-K.**, and R. Toumi, 2023: Has there been a recent shallowing of tropical cyclones? *EGU General Assembly 2023*, Vienna, Austria. (Oral)
- **Lai, T.-K.**, and R. Toumi, 2022: Static stability and declining inner-core rain rate of tropical cyclone in the last two decades. *AMS 35th Conference on Hurricanes and Tropical Meteorology*, virtual. (Oral)
- **Lai, T.-K.**, E. A. Hendricks, and M. K. Yau, 2022: Long-term effect of barotropic instability across the moat in double-eyewall tropical cyclone-like vortices. *International Conference on Heavy Rainfall and Tropical Cyclone in East Asia, T-PARCII and ISEE, Nagoya University*, virtual. (Oral)
- **Lai, T.-K.**, E. A. Hendricks, and M. K. Yau, 2021: Long-term effect of barotropic instability across the moat in double-eyewall tropical cyclone-like vortices in forced and unforced shallow-water models. *AGU Fall Meeting 2021*, virtual. (eLightning)
- **Lai, T.-K.**, E. A. Hendricks, M. K. Yau, and K. Menelaou, 2021: Roles of barotropic instability across the moat in tropical cyclone eyewall replacement cycles. *AMS 34th Conference on Hurricanes and Tropical Meteorology*, virtual. (Oral)

Professional Training

2021: Trustworthy Artificial Intelligence for Environmental Science (TAI4ES) Virtual Summer School, NCAR and National Science Foundation (NSF) AI Institute for Research on Trustworthy AI in Weather, Climate, and Coastal Oceanography (AI2ES), USA

2013: University of Reading Forecasting Module, Met Office, UK

Professional Service

2022: Member of the Working Group for Structure Change Processes: Inner Core, 10th International Workshop on Tropical Cyclones, World Meteorological Organization (WMO)

Journal Reviewer: *Journal of the Atmospheric Sciences* (AMS), *Monthly Weather Review* (AMS), *Quarterly Journal of the Royal Meteorological Society* (RMetS), *Weather* (RMetS), *Natural Hazards* (Springer), *Weather and Climate Extremes* (ScienceDirect), *Atmosphere* (MDPI), *Meteorology* (MDPI), *Scientific Online Letters on the Atmosphere* (Met. Soc. Japan)

Proposal Reviewer: NASA Postdoctoral Program

Professional Affiliations

American Meteorological Society, USA

Canadian Meteorological and Oceanographic Society, Canada

Selected Other Experience

Founder and President (2010–2011)

Meteorological Society, the Chinese University of Hong Kong

Hong Kong